PROPOSITION APPLICATION H0 IN ALGEBRAIZATION.TEX

THE STACKS PROJECT AUTHORS

Editorial note. All mathematical statement and proof text is reproduced verbatim from the author's chapter algebraization.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the wrapper class, the xr-hyper wiring and the \externaldocument list are omitted because the assembler supplies the class.

Editorial note. Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

Editorial note. The candidate is from the extractor's stacks-candidates line, not from the earlier judged initial_corpus line; the H2 judgement and its reason below were made in this module by reading the author's own statement and complete proof in the fixed checkout. Original candidate key: stacks:algebraization.tex:proposition-application-H0.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

Source reference. Original lemma 0DX3: sources/local-cohomology.tex, source lines 3078–3078. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00IP: sources/algebra.tex, source lines 12241–12241. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0BK0: sources/local-cohomology.tex, source lines 1507–1507. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0AW0: sources/dualizing.tex, source lines 1895–1895. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Source reference. Original lemma 0EIF: sources/algebraization.tex, source lines 3769–3769. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00LX: sources/algebra.tex, source lines 17905–17905. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0H48: sources/algebraization.tex, source lines 3674–3674. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Proposition 0.1. *Let $I \subset \mathfrak{a}$ be ideals of a Noetherian ring A. Let $\mathcal{F}$ be a coherent module on $U = \mathrm{Spec}(A) \setminus V(\mathfrak{a})$. Assume*

1. *A is I-adically complete and has a dualizing complex,*
2. *if $x \in Ass(\mathcal{F})$, $x \notin V(I)$, $\overline{\{x\}} \cap V(I) \not\subset V(\mathfrak{a})$, and $z \in \overline{\{x\}} \cap V(\mathfrak{a})$, then $\dim(\mathcal{O}_{\overline{\{x\}},z}) > cd(A, I) + 1$.*

Then we obtain an isomorphism

$$\mathrm{colim}\, H^0(V, \mathcal{F}) \longrightarrow \lim H^0(U, \mathcal{F}/I^n\mathcal{F})$$

where the colimit is over opens $V \subset U$ containing $U \cap V(I)$.

Proof. Let $T \subset U$ be the set of points x with $\overline{\{x\}} \cap V(I) \subset V(\mathfrak{a})$. Let $\mathcal{F} \to \mathcal{F}'$ be the surjection of coherent modules on U constructed in Local Cohomology, Lemma 0DX3. Since $\mathcal{F} \to \mathcal{F}'$ is an isomorphism over an open $V \subset U$ containing $U \cap V(I)$ it suffices to prove the lemma with $\mathcal{F}$ replaced by $\mathcal{F}'$. Hence we may and do assume for $x \in U$ with $\overline{\{x\}} \cap V(I) \subset V(\mathfrak{a})$ we have $\mathrm{depth}(\mathcal{F}_x) \geq 1$.

Let $\mathcal{V}$ be the set of open subschemes $V \subset U$ containing $U \cap V(I)$ ordered by reverse inclusion. This is a directed set. We first claim that

$$\mathcal{F}(V) \longrightarrow \lim H^0(U, \mathcal{F}/I^n\mathcal{F})$$

is injective for any $V \in \mathcal{F}$ (and in particular the map of the lemma is injective). Namely, an associated point x of $\mathcal{F}$ must have $\overline{\{x\}} \cap U \cap Y \neq \emptyset$ by the previous paragraph. If $y \in \overline{\{x\}} \cap U \cap Y$ then $\mathcal{F}_x$ is a localization of $\mathcal{F}_y$ and $\mathcal{F}_y \subset \lim \mathcal{F}_y/I^n\mathcal{F}_y$ by Krull's intersection theorem (Algebra, Lemma 00IP). This proves the claim as a section $s \in \mathcal{F}(V)$ in the kernel would have to have empty support, hence would have to be zero.

Choose a finite A-module M such that $\mathcal{F}$ is the restriction of $\widetilde{M}$ to U, see Local Cohomology, Lemma 0BK0. We may and do assume that $H^0_{\mathfrak{a}}(M) = 0$. Let $\mathrm{Ass}(M) \setminus V(I) = \{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$. We will prove the lemma by induction on n. After reordering we may assume that $\mathfrak{p}_n$ is a minimal element of the set $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$ with respect to inclusion, i.e, $\mathfrak{p}_n$ is a generic point of the support of M. Set

$$M' = H^0_{\mathfrak{p}_1 \ldots \mathfrak{p}_{n-1} I}(M)$$

and $M'' = M/M'$. Let $\mathcal{F}'$ and $\mathcal{F}''$ be the coherent $\mathcal{O}_U$-modules corresponding to M' and M''. Dualizing Complexes, Lemma 0AW0 implies that M'' has only one associated prime, namely $\mathfrak{p}_n$. Hence $\mathcal{F}''$ has only one associated point and we see that condition (3)(a) of Lemma 0EIF holds; thus the map $\mathrm{colim}\, H^0(V, \mathcal{F}'') \to$

$\lim H^0(U, \mathcal{F}''/I^n\mathcal{F}'')$ is an isomorphism. On the other hand, since $\mathfrak{p}_n \notin V(\mathfrak{p}_1 \ldots \mathfrak{p}_{n-1}I)$ we see that $\mathfrak{p}_n$ is not an associated prime of M'. Hence the induction hypothesis applies to M'; note that since $\mathcal{F}' \subset \mathcal{F}$ the condition $\text{depth}(\mathcal{F}'_x) \geq 1$ at points x with $\overline{\{x\}} \cap V(I) \subset V(\mathfrak{a})$ holds, see Algebra, Lemma 00LX. Thus the map $\text{colim}\, H^0(V, \mathcal{F}') \to \lim H^0(U, \mathcal{F}'/I^n\mathcal{F}')$ is an isomorphism too. We conclude by Lemma 0H48. $\square$